

Colonialism as a Natural Experiment

Week 3

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Today

1. **The problem.** Institutions are the leading answer to “why there?” — and the hardest thing in economics to measure the effect of.
2. **Two famous designs.** Acemoglu, Johnson and Robinson instrument with settler mortality. Dell uses a colonial boundary.
3. **The audit.** Conley and Kelly ask whether either result survives standard errors that know the data are spatial.

Week 2 gave you the toolkit and the ethic. Today is the first time we point them at a literature and pull the trigger.

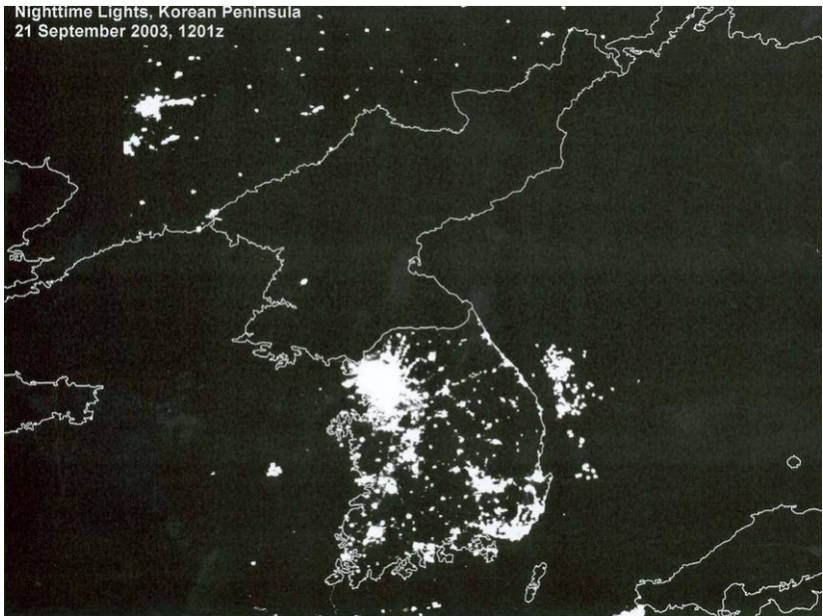
The question we left hanging

Why there, and why then?

Week 1 ended with a fact and a question. Incomes diverged after 1800, and they diverged **unevenly**. Three candidate explanations have done most of the work since.

- **Geography.** Climate, disease, soil, access to the sea.
- **Culture.** Beliefs, trust, family structure, religion.
- **Institutions.** The rules that decide who may keep what they produce.

Was colonization good or bad for the colonies? Why did some colonies end up rich and others poor? Can something that ended two centuries ago still be moving incomes today? And what would make a good natural experiment here?



Nighttime lights over the Korean peninsula, 21 September 2003 (NASA/DMSP). One peninsula, one people, one border drawn in 1945. Institutions do seem to matter.

Institutions

Institutions

The formal and informal rules that structure social interaction. In this literature the working measure is usually narrower — protection against expropriation, constraints on the executive, security of property.

- Rich countries have better institutions. Nobody disputes the correlation.
- But which way does it run? Good institutions may cause growth. Growth may buy good institutions. Or some third thing — culture, geography, human capital — may drive both.

→ give a slightly more complicated definition...

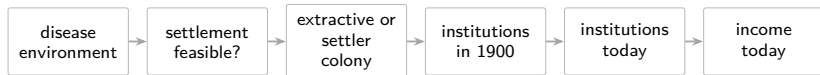
What would count as a natural experiment

- We cannot randomize institutions across countries. So we look for a moment in history that did something close to it **for reasons unrelated to the outcome we care about**.
- Colonial history is a tempting place to look. One shock, imposed from outside, on hundreds of societies, with wildly different rules imposed in different places.
- The whole question is always the same. **Why did the colonizer treat this place differently from that one** — and is that reason plausibly unrelated to how rich the place would be today?

Snow's three water companies served the same streets, and households had not chosen between them. That is the standard. Everything today is an attempt to find a colonial version of it.

Settler mortality as an instrument

The Acemoglu, Johnson and Robinson chain



- Where Europeans could not survive, they did not settle. Where they did not settle, they built machinery for **extraction** rather than for production.
- Those arrangements outlived the colonizers, and the claim is that they are still visible in the rules countries live under now.

Acemoglu, Johnson and Robinson (2001), "The Colonial Origins of Comparative Development."

Two kinds of colony

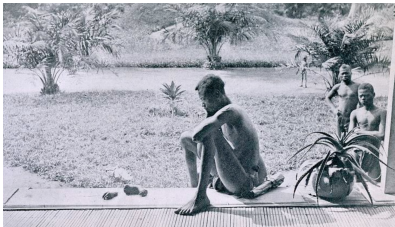
Extractive

- The Belgian Congo, most of Spanish America.
- Purpose is to move resources to the metropole. Forced labor, monopolies, no protection for local property.
- Few Europeans ever live there.

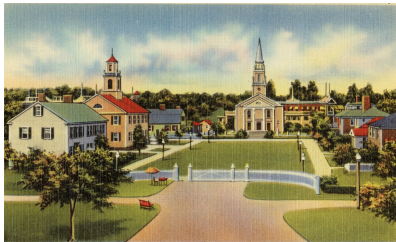
Settler/Inclusive

- The United States, Canada, Australia, New Zealand.
- Europeans arrive to stay, and demand the rules they had at home — trial by jury, elected legislatures, protection from arbitrary seizure.
- They are writing rules they expect to live under.

The mechanism is that a settler has to live under the rules he writes, and an administrator collecting rubber does not.



Congo Free State, 1904. Nsala of Wala with the hand and foot of his daughter, the punishment for a short rubber quota. Photograph by Alice Seeley Harris.



A New England town common — meetinghouse, church, school. Settlers writing rules they expected to live under.

Why the obvious regression will not do

Run income today on institutions today across countries and you get a large, precise coefficient. Acemoglu, Johnson and Robinson report **0.52** in their base sample of 64 countries, with institutions accounting for over half the variation in income.

- **Reverse causality.** Rich countries can afford courts, registries and an independent judiciary.
- **Omitted variables.** Anything that makes a country both rich and well governed sits in the error term.
- **Measurement error.** An expropriation-risk index rated by country-risk analysts is a noisy proxy for the rules that actually bind, and noise pushes the coefficient toward zero.

The instrument

What settler mortality has to do

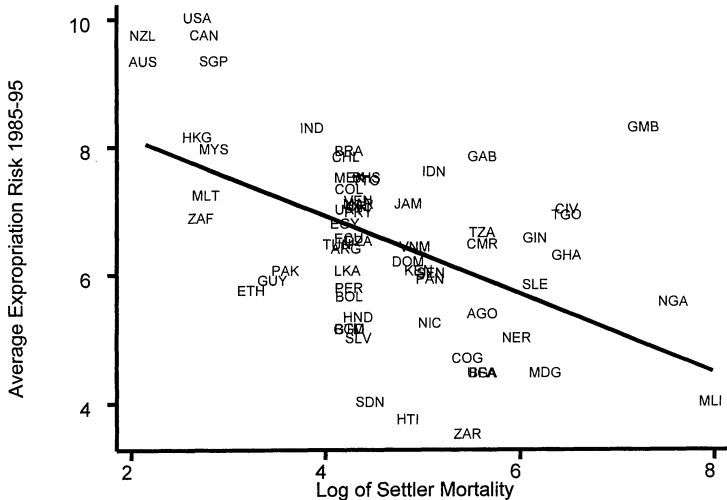
Relevance. Mortality must actually predict the institutions colonies ended up with. It does — settler mortality alone accounts for about **27 percent** of the variation in institutions across these countries today.

Exclusion. The death rate among European soldiers and bishops two centuries ago must affect income today **only** through the institutions it produced.

- Relevance you can test. **Exclusion you cannot.** It is an argument that the paper spends most of its length making.

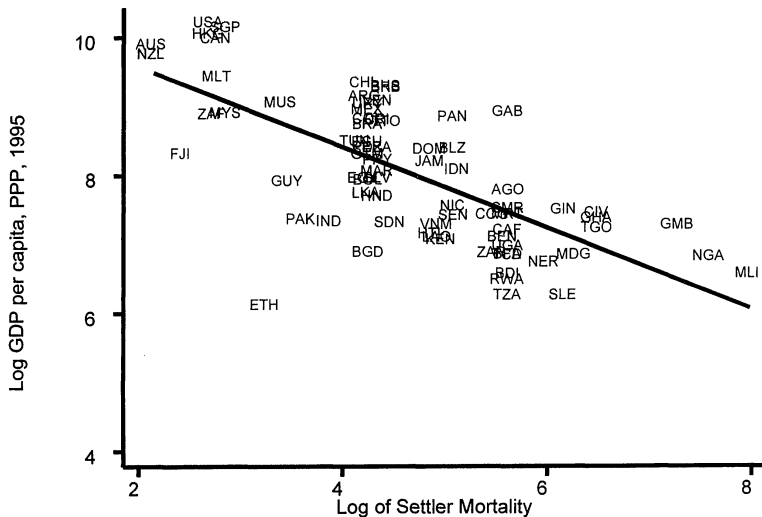
Their case is that yellow fever and malaria killed Europeans who had no immunity while leaving immune local adults largely alone, so the historical death rate is not a proxy for a disease burden that still suppresses income today.

The first stage



Acemoglu, Johnson and Robinson (2001), Figure 3. Higher settler mortality then, weaker protection against expropriation now. This is the regression Conley and Kelly will come back to at the end of class.

The reduced form



Acemoglu, Johnson and Robinson (2001), Figure 1. Income today on mortality then. This slope divided by the first stage's is the 2SLS estimate.

The two equations

Second stage — what we want to know.

$$\log y_i = \mu + \alpha R_i + X_i' \gamma + \varepsilon_i$$

First stage — where R_i comes from.

$$R_i = \zeta + \beta \log M_i + X_i' \delta + \nu_i$$

- y_i is income per head, R_i is protection against expropriation, M_i is settler mortality and X_i is the vector of controls. α is the object of interest.
- The exclusion restriction is that $\log M_i$ does not appear in the first equation. Two-hundred-year-old death rates affect income today **only** through R_i .

Everything turns on that one absence. It cannot be tested, only argued, and the argument is most of the paper.

	Base sample (1)	Base sample (2)	Base sample without Neo-Europes (3)	Base sample without Neo-Europes (4)	Base sample without Africa (5)	Base sample without Africa (6)	Base sample with continent dummies (7)	Base sample with continent dummies (8)	Base sample, dependent variable is log output per worker (9)
Panel A: Two-Stage Least Squares									
Average protection against expropriation risk 1985–1995	0.94 (0.16)	1.00 (0.22)	1.28 (0.36)	1.21 (0.35)	0.58 (0.10)	0.58 (0.12)	0.98 (0.30)	1.10 (0.46)	0.98 (0.17)
Latitude		−0.65 (1.34)		0.94 (1.46)		0.04 (0.84)		−1.20 (1.8)	
Asia dummy							−0.92 (0.40)	−1.10 (0.52)	
Africa dummy							−0.46 (0.36)	−0.44 (0.42)	
“Other” continent dummy							−0.94 (0.85)	−0.99 (1.0)	
Panel B: First Stage for Average Protection Against Expropriation Risk in 1985–1995									
Log European settler mortality	−0.61 (0.13)	−0.51 (0.14)	−0.39 (0.13)	−0.39 (0.14)	−1.20 (0.22)	−1.10 (0.24)	−0.43 (0.17)	−0.34 (0.18)	−0.63 (0.13)
Latitude		2.00 (1.34)		−0.11 (1.50)		0.99 (1.43)		2.00 (1.40)	
Asia dummy							0.33 (0.49)	0.47 (0.50)	
Africa dummy							−0.27 (0.41)	−0.26 (0.41)	
“Other” continent dummy							1.24 (0.84)	1.1 (0.84)	
R ²	0.27	0.30	0.13	0.13	0.47	0.47	0.30	0.33	0.28
Panel C: Ordinary Least Squares									
Average protection against expropriation risk 1985–1995	0.52 (0.06)	0.47 (0.06)	0.49 (0.08)	0.47 (0.07)	0.48 (0.07)	0.47 (0.07)	0.42 (0.06)	0.40 (0.06)	0.46 (0.06)
Number of observations	64	64	60	60	37	37	64	64	61

Acemoglu, Johnson and Robinson (2001), Table 4. Panel A is 2SLS, Panel B the first stage, Panel C OLS. Read column (1) downward.

The whole result, before any regression

TABLE 1—DESCRIPTIVE STATISTICS

	Whole world	Base sample	By quartiles of mortality			
			(1)	(2)	(3)	(4)
Log GDP per capita (PPP) in 1995	8.3 (1.1)	8.05 (1.1)	8.9	8.4	7.73	7.2
Log output per worker in 1988 (with level of United States normalized to 1)	−1.70 (1.1)	−1.93 (1.0)	−1.03	−1.46	−2.20	−3.03
Average protection against expropriation risk, 1985–1995	7 (1.8)	6.5 (1.5)	7.9	6.5	6	5.9
Constraint on executive in 1990	3.6 (2.3)	4 (2.3)	5.3	5.1	3.3	2.3
Constraint on executive in 1900	1.9 (1.8)	2.3 (2.1)	3.7	3.4	1.1	1
Constraint on executive in first year of independence	3.6 (2.4)	3.3 (2.4)	4.8	2.4	3.1	3.4
Democracy in 1900	1.1 (2.6)	1.6 (3.0)	3.9	2.8	0.19	0
European settlements in 1900	0.31 (0.4)	0.16 (0.3)	0.32	0.26	0.08	0.005
Log European settler mortality	n.a.	4.7 (1.1)	3.0	4.3	4.9	6.3
Number of observations	163	64	14	18	17	15

Acemoglu, Johnson and Robinson (2001), Table 1. Standard deviations in parentheses.

Reading that table

Panel	What it estimates	Column (1)
C, OLS	the naive association	0.52 (0.06)
B, first stage	institutions on mortality	-0.61 (0.13), $R^2 = 0.27$
A, 2SLS	the causal estimate	0.94 (0.16)

- The first stage is strong and correctly signed. Higher mortality then, worse institutions now.
- **2SLS is nearly double OLS.**
- Their reading is that **measurement error** in an expropriation-risk index dominates and attenuation was doing more damage than endogeneity. Read the rest of the columns as robustness. Dropping the neo-Europes gives 1.28; dropping Africa gives 0.58.

Why a noisy regressor pulls the slope toward zero

Classical measurement error

We want α in $\log y_i = \mu + \alpha R_i^* + \varepsilon_i$, where R_i^* is the protection a country actually offers. What the data hand us is $R_i = R_i^* + u_i$, with the noise u_i unrelated to R_i^* and to ε_i .

$$\text{plim } \hat{\alpha}_{\text{OLS}} = \alpha \frac{\sigma_{R^*}^2}{\sigma_{R^*}^2 + \sigma_u^2} = \lambda \alpha, \quad 0 < \lambda < 1$$

- A slope is a covariance over a variance. The noise enlarges the denominator and adds nothing to the numerator, so the estimate arrives shrunk toward zero. That shrinkage is **attenuation bias**, and λ is the **reliability ratio**, the signal share of the observed variance.
- An instrument correlated with R_i^* and unrelated to u_i recovers α . That is how 2SLS can land **above** OLS.

Take the gap at face value and $\lambda \approx 0.52/0.94 \approx 0.55$. Barely half the variance in a country-risk rating would be signal. This is the classical case; let raters anchor on how rich a country already looks and the error correlates with R^* , and the bias can run either way.

What the instrument buys

	OLS	2SLS
Effect of institutions on log income	0.52	0.94
Standard error	(0.06)	(0.16)
Adding latitude		1.00 (0.22)
Excluding the neo-Europes		1.28

- The instrumented estimate is **larger** than OLS.
- Their reading is that attenuation from measurement error was doing more damage than reverse causality and omitted variables combined.
- Dropping the United States, Canada, Australia and New Zealand does not break it, which answers the first objection most people raise.

Acemoglu, Johnson and Robinson (2001), Tables 2 and 4, base sample of 64 countries.

Reading a difference in logs

The conversion

When two countries differ by Δ in log income, the richer one is e^Δ times as rich, where $e \approx 2.7$.

Log gap	Times as rich
0.05	1.05
0.10	1.11
0.50	1.65
1.00	2.72
2.06	7.85
2.53	12.55

- Under about 0.2 a log gap reads straight off as a percentage, so 0.05 in logs is close to 5 percent richer.
- Past that the two part company and you have to exponentiate. A log gap of 1 is a factor of e itself, about **2.7** times as rich, which is 170 percent more rather than 100.

We work in logs because differences add and proportional changes come out symmetric. The price is that the numbers stop being readable, so convert before you judge a magnitude.

Is that a big number?

Nigeria sits near the 25th percentile of the institutions index at **5.6**, Chile near the 75th at **7.8**. The gap is **2.2** points.

$$\underbrace{0.94}_{\text{2SLS estimate}} \times \underbrace{2.2}_{\text{gap in the index}} = \underbrace{2.06}_{\text{predicted log gap}}$$

	In its own units	In SDs	Times as rich
Institutions, Nigeria to Chile	2.2 points	1.5	—
Income gap the design predicts	2.06 logs	1.9	7.8
Income gap actually observed	2.53 logs	2.3	12.6

Standard deviations from AJR Table 1, base sample — 1.5 for the index, 1.1 for log income.

One standard deviation of institutions buys **1.28 standard deviations** of log income, and the design hands institutions about four fifths of the Nigeria–Chile gap. AJR call that an upper bound, since some of the measured institutional gap is noise.

Why that is the question to ask

- **A coefficient is not a magnitude until it has units.** Institutions are an index nobody has intuition for, and income is in logs. Neither number means anything until you put both on a common scale.
- **Standard deviations are the common scale.** They ask how far a country would have to move within the observed spread, which is a question you can argue about.
- **And 1.28 is the number to argue about.** Moving one standard deviation along an index of expropriation risk would move income more than a standard deviation. Ask whether any actual reform has ever done that.
- **To picture one standard deviation,** it runs about two thirds of the way from Nigeria to Chile on this index. So the claim is that two thirds of that institutional distance is worth more than a full standard deviation of income.

This is the McCloskey question from Week 2. How much of the thing we are trying to explain does it explain?

Critiques?

- **The disease environment never left.** Malaria still suppresses income in some of these places. If so, mortality reaches income by a second route and the exclusion restriction fails.
- **Mortality proxies geography.** Latitude, climate and coastal access all correlate with where Europeans died.
- **Human capital.** Settlers brought schooling and skills as well as rules. Glaeser and co-authors (2004) argue that this is the channel that matters.
- **The data.** Where did the mortality numbers actually come from? Albouy (2012) went and checked. in a moment.

Every one of these is an argument about the **exclusion restriction**. That is what a debate over an IV paper always turns out to be about.

How fast can institutions change?



Hosni Mubarak, president of Egypt for nearly thirty years, and the same man in a Cairo courtroom cage in August 2011, six months after eighteen days of protest ended his rule.

Hold this pair of photographs against a design that needs rules from 1600 to still bind in 1995.

The persistence critique

The chain has a middle link. Mortality shapes the institutions of 1900, and those institutions have to reach 1995 intact enough to carry the whole effect.

- Egypt dismantled a thirty-year presidency in eighteen days. Rules can change fast, and they do. So persistence is a thing to be explained.
- AJR offer three mechanisms. Building constraining institutions is costly and the colonizers either sank the cost or did not. A small elite keeps more per member by staying extractive. And people who invested under the old rules will defend them.
- The older worries still apply — credible commitment to reform is hard, and inequality tends to reproduce the rules that produced it.

THE COLONIAL ORIGINS OF COMPARATIVE DEVELOPMENT: AN INVESTIGATION OF THE SETTLER MORTALITY DATA

David Y. Albouy

ABSTRACT

In a seminal contribution, Acemoglu, Johnson, and Robinson (2001) argue property-rights institutions powerfully affect national income, using estimated mortality rates of early European settlers to instrument capital expropriation risk. However 36 of the 64 countries in their sample are assigned mortality rates from other countries, typically based on mistaken or conflicting evidence. Also, incomparable mortality rates from populations of laborers, bishops, and soldiers - often on campaign - are combined in a manner favoring their hypothesis. When these data issues are controlled for, the relationship between mortality and expropriation risk lacks robustness, and instrumental-variable estimates become unreliable, often with infinite confidence intervals.

Albouy (2012), American Economic Review. The number to take from the abstract is 36 of 64 — more than half the sample carries a mortality rate assigned from some other country.

Where the numbers came from

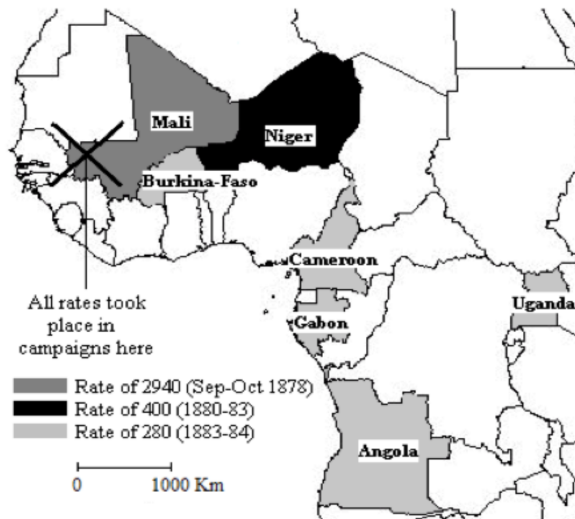


FIGURE 1: ASSIGNMENT OF MORTALITY RATES FROM MALI

Albouy (2012), Figure 1. Every rate shown was measured during military campaigns in one corner of Mali in 1878–84, then assigned to Niger, Burkina Faso, Cameroon, Gabon, Uganda and Angola.

What the first stage looks like afterward

Control Variables	No Controls	Latitude Control	Without Neo- Europes	Continent Dummies	Continent Dummies & Latitude	Mean Temp and Min Rain	Percent European, 1975	Malaria in 1994
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Panel E: Removing conjectured rates, correcting Mali, adding campaign and laborer dummies, and revising with new data (34 countries and</i>								
Log mortality (β)	-0.36	-0.22	-0.10	-0.25	-0.10	0.02	-0.15	-0.14
(heteroscedastic s.e.)	(0.22)	(0.24)	(0.21)	(0.25)	(0.24)	(0.24)	(0.23)	(0.27)
<i>p</i> -value of log mortality	0.11	0.35	0.66	0.32	0.69	0.93	0.53	0.61
<i>p</i> -value of dummies	0.01	0.02	0.02	0.28	0.30	0.00	0.06	0.10
<i>p</i> -value of controls	-	0.11	-	0.14	0.15	0.001	0.04	0.03

Albouy (2012), Table 1, Panel E.

- Remove the conjectured rates, correct Mali, add campaign and laborer controls, and the first stage goes from -0.61 (0.13) to -0.36 (0.22), $p = 0.11$. Add latitude and it is -0.22 , $p = 0.35$.
- Acemoglu, Johnson and Robinson replied in the same issue and defended the assignments. Treat it as a **live dispute**.

Week 2's ethic again. The data series is part of the design, and shoe leather cuts both ways.

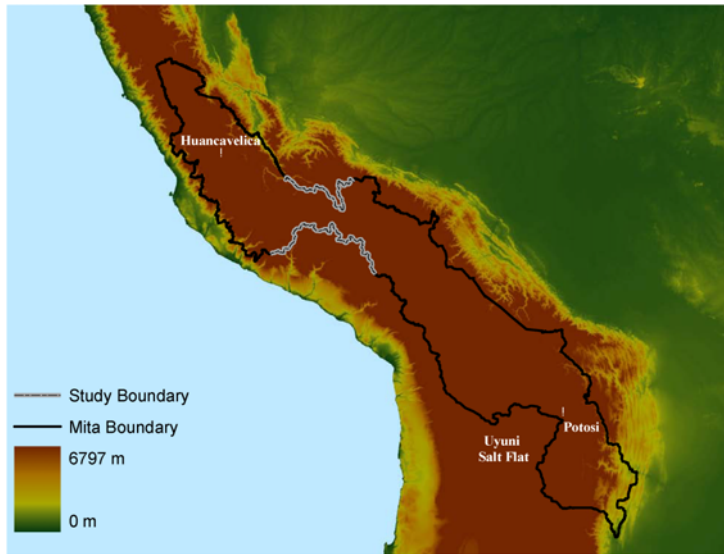
The Peruvian Mining Mita

The mita

- A forced-labor system run by the Spanish crown in Peru and Bolivia from **1573 to 1812**.
- Over **200 indigenous communities** were required to send **one seventh** of their adult men to the silver mines at Potosí and the mercury mines at Huancavelica.
- Conscripts traveled as far as 1,000 kilometers each way and were paid well below subsistence.
- Districts inside the boundary sent the same share of their population. Districts outside sent **nobody**. The obligation changed **discretely** at a line on a map.

Dell (2010), "The Persistent Effects of Peru's Mining Mita."

The mita boundary



Dell (2010), Figure 1. The mita boundary in black, the study boundary in light gray, elevation behind. She uses only the southern segment that transects the Andean range.

Why a boundary is worth more than a country

The identifying assumptions

Dell names two. First, everything relevant except treatment varies **smoothly** across the boundary, so districts just inside and just outside are a fair comparison and the jump at the line is the effect of the mita.

Second, the RD polynomial has to approximate that smooth variation well, so that a bend in the surface is not mistaken for a jump.

- Assignment was made on two stated criteria — **distance to the mines** and **elevation**, since Spanish officials believed only highland people could survive work at 4,000 meters.
- Both criteria are measurable, and both vary smoothly through space. That is what makes the discontinuity usable rather than suspicious.

Compare the samples. Acemoglu, Johnson and Robinson compare Nigeria with Chile. Dell compares a village with the village over the ridge.

The shoe leather

Dell does not assert the boundary is clean. She goes and checks.

- She discards most of the boundary and keeps only the southern segment, because elsewhere the line follows the Andean precipice — so elevation and ethnic composition change discontinuously and the design would be dead on arrival.
- On the segment she keeps, elevation, ethnicity and other observables are statistically indistinguishable across the line.
- She then goes to census documents from **just before 1573** — local tribute rates, allocation of tribute revenue, demography — and finds no differences across the boundary before the mita existed.

Evidence that treatment and control were alike **before treatment**, collected from sixteenth-century tax records.

What she finds, four centuries later

- Household consumption is about **25 percent lower** in districts that were subject to the mita.
- Childhood stunting is about **6 percentage points higher**, against a baseline of roughly 40 percent stunting across the region.
- Mita districts are less connected to the road network, and their residents are considerably more likely to be subsistence farmers.

The mita was abolished in 1812. These are households measured in 2001 and children measured in 2005. The institution has been gone for the better part of two centuries and the gap is still there.

Dell (2010), Table II. Stunting estimates run from 0.055 with a standard error of 0.030 upward across specifications.

The balance table behind that claim

	Sample Falls Within								
	<100 km of <i>Mita</i> Boundary			<75 km of <i>Mita</i> Boundary			<50 km of <i>Mita</i> Boundary		
	Inside	Outside	s.e.	Inside	Outside	s.e.	Inside	Outside	s.e.
GIS Measures									
Elevation	4042	4018	[188.77] (85.54)	4085	4103	[166.92] (82.75)	4117	4096	[169.45] (89.61)
Slope	5.54	7.21	[0.88]* (0.49)***	5.75	7.02	[0.86] (0.52)**	5.87	6.95	[0.95] (0.58)*
Observations	177	95		144	86		104	73	
% Indigenous	63.59	58.84	[11.19] (9.76)	71.00	64.55	[8.04] (8.14)	71.01	64.54	[8.42] (8.43)
Observations	1112	366		831	330		683	330	
Log 1572 tribute rate	1.57	1.60	[0.04] (0.03)	1.57	1.60	[0.04] (0.03)	1.58	1.61	[0.05] (0.04)

Dell (2010), Table I. Brackets are Conley spatial standard errors, parentheses robust ones. The <25 km window is cut for space.

- **Elevation balances** everywhere, and it was one of the two assignment criteria. So does % indigenous, and so does the **1572 tribute rate**, a year before the mita began.
- **Slope does not**, 5.54 inside against 7.21 outside, which is why it sits in her controls on the next slide.

Look at how slope is starred. Three stars on the robust standard error, one on Conley's. Dell is already using spatial standard errors in 2010.

Her specification

$$c_{idb} = \alpha + \gamma \text{mita}_d + X_{id}\beta + f(\text{geographic location}_d) + \phi_b + \varepsilon_{idb}$$

- γ is the mita effect. ϕ_b are fixed effects for whichever of the **boundary segments** an observation lies nearest to, so the comparison is always local.
- X_{id} carries elevation and slope, plus household demographics.
- $f(\cdot)$ is the **RD polynomial**, and she runs three versions of it — a cubic in latitude and longitude, a cubic in distance to Potosí, and a cubic in distance to the boundary itself.
- Standard errors are clustered by district. Metropolitan Cusco is dropped throughout, since it was the Inca capital and its prosperity is not about the mita.

Running three polynomials is not indecision. The RD estimate is only as good as the way you control for smooth geography, so she shows you what happens under each and lets you see the disagreement.

And the disagreement is worth seeing

TABLE II
LIVING STANDARDS^a

Sample Within:	Dependent Variable						
	Log Equiv. Household Consumption (2001)			Stunted Growth, Children 6–9 (2005)			
	<100 km of Bound. (1)	<75 km of Bound. (2)	<50 km of Bound. (3)	<100 km of Bound. (4)	<75 km of Bound. (5)	<50 km of Bound. (6)	Border District (7)
Panel A. Cubic Polynomial in Latitude and Longitude							
<i>Mita</i>	−0.284 (0.198)	−0.216 (0.207)	−0.331 (0.219)	0.070 (0.043)	0.084* (0.046)	0.087* (0.048)	0.114** (0.049)
R^2	0.060	0.060	0.069	0.051	0.020	0.017	0.050
Panel B. Cubic Polynomial in Distance to Potosí							
<i>Mita</i>	−0.337*** (0.087)	−0.307*** (0.101)	−0.329*** (0.096)	0.080*** (0.021)	0.078*** (0.022)	0.078*** (0.024)	0.063* (0.032)
R^2	0.046	0.036	0.047	0.049	0.017	0.013	0.047
Panel C. Cubic Polynomial in Distance to <i>Mita</i> Boundary							
<i>Mita</i>	−0.277*** (0.078)	−0.230** (0.089)	−0.224** (0.092)	0.073*** (0.023)	0.061*** (0.022)	0.064*** (0.023)	0.055* (0.030)
R^2	0.044	0.042	0.040	0.040	0.015	0.013	0.043
Geo. controls	yes	yes	yes	yes	yes	yes	yes
Boundary F.E.s	yes	yes	yes	yes	yes	yes	yes
Clusters	71	60	52	289	239	185	63
Observations	1478	1161	1013	158,848	115,761	100,446	37,421

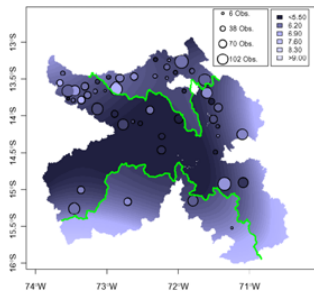
Dell (2010), Table II. Consumption in columns 1–3, childhood stunting in 4–7, with the sample narrowing to within 100, 75 and 50 kilometers of the boundary as you move right and column 7 restricted to districts on the boundary itself. Three panels, three RD polynomials.

What that table admits

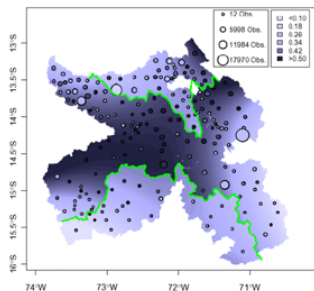
- Panels B and C, the one-dimensional polynomials, give consumption effects of -0.22 to -0.34 in logs, all significant at 1 or 5 percent. That is the “about 25 percent lower” headline.
- **Panel A does not.** With a flexible cubic in latitude and longitude, the point estimates are similar — -0.28 , -0.22 , -0.33 — but the standard errors more than double and none of the three consumption estimates is significant.
- Dell says so plainly. She gives three reasons — the flexibility of the specification, the small number of observations and clusters, and measurement error in consumption. Only the first is about spatial trend, and it is the one worth carrying into the last hour.

Hold that thought for the last hour. Panel A adds a **flexible two-dimensional spatial control** and the result loses significance.

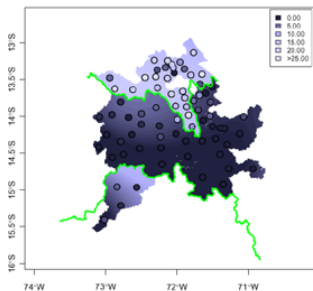
Conley and Kelly are about to do exactly that, to the whole literature.



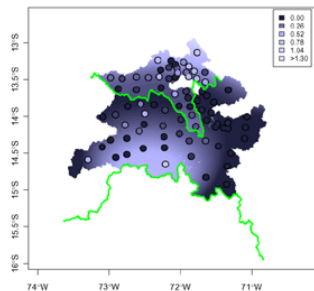
(a) Consumption (2001)



(b) Stunting (2005)

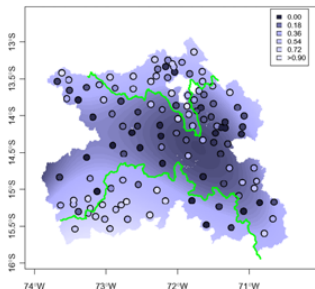


(c) *Haciendas* (1689)

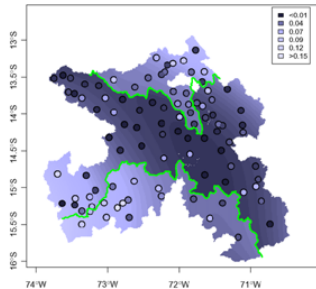


(d) *Haciendas* (1845)

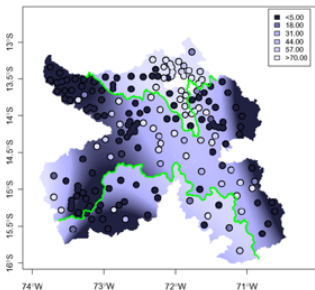
Dell (2010), Figure 2, panels (a) to (d). Dots are districts, the background is the fitted surface from the cubic in latitude and longitude, and the green line is the mita boundary. RD plots in three dimensions.



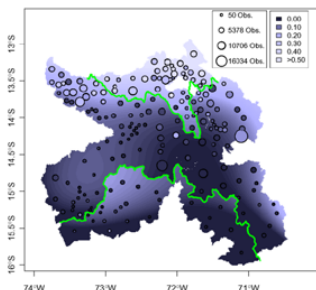
(e) Haciendas (1940)



(f) Education (1876)



(g) Road Density (2006)



(h) Ag. Market Participation (1994)

Dell (2010), Figure 2, panels (e) to (h). The mechanism, mapped. Haciendas in 1940, schooling in 1876, road density in 2006 and agricultural market participation in 1994, all of them changing at the same line.

The mechanism, which is the real contribution

Finding persistence is the easy part. Dell asks **through what**.

- **Land tenure.** The crown blocked **haciendas** — large estates with attached labor — inside mita districts, to avoid competing for the labor it was conscripting. Outside, haciendas formed. The difference is still visible in 1689, 1845 and 1940.
- **Public goods.** Hacienda owners were a landed elite with the standing to lobby for roads and the means to organize their construction. Communal villages had no such patron.
- **Education.** Schooling was historically lower inside the mita.

The uncomfortable result is that the **large landowners** are the ones associated with better outcomes today. Persistence runs through channels that do not flatter anybody's politics, which is a good sign that it was measured rather than assumed.

Two designs, side by side

	Acemoglu et al.	Dell
Design	instrumental variable	boundary discontinuity
Assignment	disease environment	distance and elevation
Compares	Nigeria with Chile	village with next village
Sample	64 countries	districts along one line
Answers	do institutions matter?	through what do they persist?
Weak point	the exclusion restriction	does it travel anywhere else?

- Dell buys a much more credible comparison and pays for it in **external validity**. Nothing about Peruvian silver districts tells you about Ghana.
- That trade is the central one in empirical economics, and you should be able to state which side of it any paper sits on.

The standard errors of persistence

An uncomfortable fact about this literature

"All of these objections crumble into irrelevance in the face of one blunt fact — the unusual explanatory power of these persistence variables. While a judicious choice of variables or time periods might coax a t statistic past 1.96, there would appear to be no way that the t statistics of four, five, or even larger that appear routinely in this literature could be the result of massaging regressions."

- Conley and Kelly grant the point, then turn it around. If the t statistics are too large to have been manufactured, perhaps they are too large to be **real**.
- Most of these papers report p-values on the order of **0.001 or smaller**.

Conley and Kelly (2025), "The Standard Errors of Persistence."

Persistence regressions are spatial regressions

- Both sides of these regressions are **maps**. Settler mortality is a map. Income today is a map. The mita boundary is a map.
- Maps are smooth. Neighboring places resemble each other, on the outcome and on the treatment alike.
- So the effective number of independent observations is far smaller than the number of rows in the data set, and a standard error computed as though the rows were independent is **far too small**.

Two tests and a repair

- **The placebo test.** Replace the real treatment with spatial noise — a variable with the same smoothness and no possible relationship to the outcome. Run the study's own regression and inference. How often does pure noise come out significant?
- **The synthetic outcomes test.** Generate outcomes from a trend plus spatial noise, independent of treatment by construction and ask whether the real coefficient stands out against that reference distribution.
- **The repair.** Add a low-dimensional spatial basis to soak up the trend then use large-cluster standard errors. Their R package `spatInfer` does it.

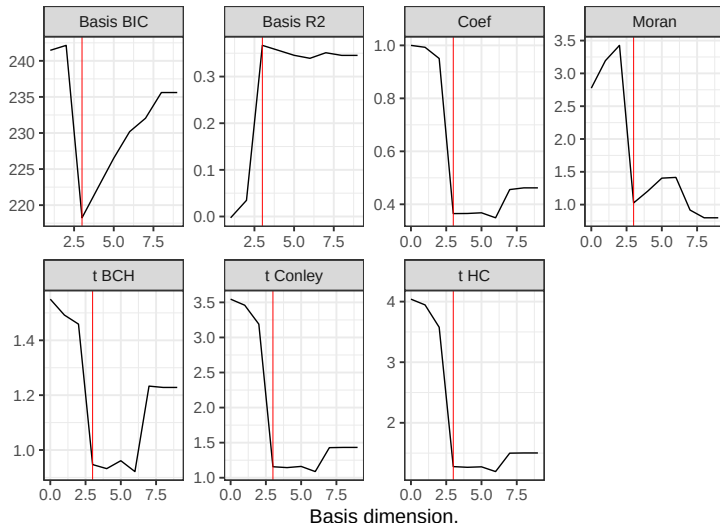
What their procedure actually does

1. **Build a spatial basis.** Lay a tensor spline over the map — think of four undulating threads running north to south crossed with four running east to west — and take the first few **principal components** of those terms, with the number chosen by BIC.
2. **Add them to the study's own regression.** They soak up the smooth spatial trend, which is what makes two unrelated maps correlate.
3. **Fix the standard errors.** Use a small number of large clusters, chosen by k-medoids, rather than many small ones. Neighboring small clusters are correlated with each other, so many small clusters understate uncertainty.
4. **Then test.** The placebo test replaces the treatment with spatial noise; the synthetic outcomes test replaces the outcome with trend plus spatial noise.

Note that step one is Dell's Panel A, generalized. She controlled for smooth geography with a cubic in latitude and longitude and lost significance. Conley and Kelly do the same thing more flexibly and to thirty papers at once.

What it does to Acemoglu, Johnson and Robinson

Acemoglu, Colonial. Obs=64, Splines=3, Basis=3, BCH Clusters=4.



Conley and Kelly (2025), Figure 6, applied to the first stage we looked at earlier. Moran is a z statistic for spatial correlation in the residuals.

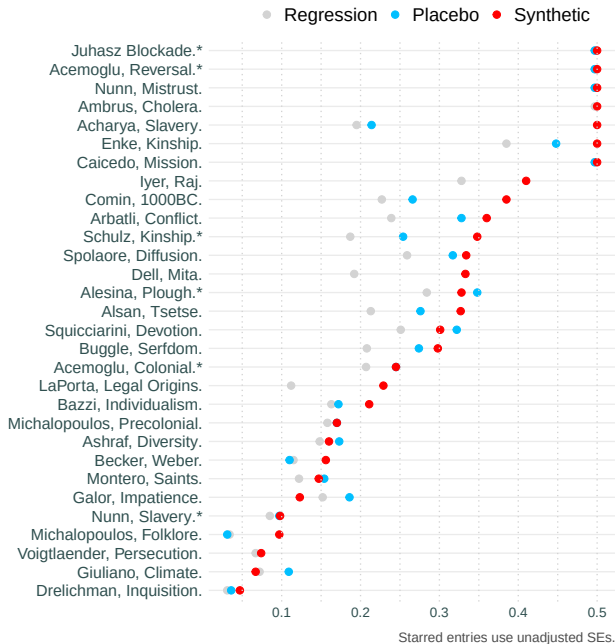
Reading those seven panels

Every panel shares one horizontal axis, the **basis dimension**, which is how much smooth spatial structure has been taken out. The red line is where BIC says to stop, here at three.

Panel	What it is	What it does here
Basis BIC	fit criterion for the basis	bottoms out at 3, which sets the red line
Basis R2	what the basis alone explains	0 to about 0.35
Coef	AJR's first-stage slope	1.00 down to about 0.36
Moran	z statistic for spatial correlation left in the residuals	3.4 down to about 1.0
t BCH, t Conley, t HC	three ways of computing the standard error	1.6, 3.5 and 4.1, all near 1 by 3

Look at the three t statistics at basis dimension **zero**, before anything is added. HC says 4.1, Conley 3.5, BCH 1.6. Same coefficient, same data, and the choice of standard error alone carries you from decisive to insignificant.

Then pause on Basis R2. Three principal components of a 3×3 spline, carrying no economics whatever, account for 35 percent of the variation in institutions across these 64 countries. That is Conley and Kelly's own figure.



Conley and Kelly (2025), Figure 7. Grey dots are each study's own regression re-run with the spatial basis. Starred rows use HC inference rather than clustered.

Reading the audit

- One row per study, thirty of them, ordered by significance. The horizontal axis is a **p-value**, so left is stronger and the eye should travel to the bottom left.
- Three dots per row, all of them p-values against the **same null of no treatment effect**, reached three different ways.
 - Grey, Regression.** The study's own regression, re-run with the spatial basis and large-cluster standard errors.
 - Blue, Placebo.** The treatment replaced by spatial noise of the same smoothness. Left blank for binary treatments, which is why Dell has no blue dot.
 - Red, Synthetic.** The outcome replaced by trend plus spatial noise.
- Starred rows use HC standard errors, the rest use BCH.

A study survives if its grey dot sits left of **0.05**. Two of the thirty do.

The grey dot asks whether the effect survives. The blue and red ask the harder question, whether noise carrying the same spatial structure would have produced the finding anyway.

What happens to our two papers

- Across all **thirty** studies, only **two** keep a treatment effect significant at 5 percent and two more at 10 percent. Only one can reject the synthetic outcomes null.
- **Acemoglu, Johnson and Robinson.** What Conley and Kelly re-run is the **first stage** — the scatter we looked at earlier. Adding three principal components from a minimal spatial basis cuts the coefficient by more than half and drops the t statistic **from 4 to 1.5**. **Ask yourself why that is worse than losing the second stage.**
- **Dell.** The mita regression survives no better. Its p-value under their procedure is about **0.19**, and about 0.33 on the synthetic outcomes test. Note which one they took — Panel C, her **tightest** specification, the one with the smallest standard errors.

Neither paper is among the two that survive. The boundary design does not rescue Dell here, which is worth sitting with — a local comparison is still a comparison between places on a map.

Why the first stage is the worse place to lose

The first stage is not the result, so who cares if it wobbles?

- **Because everything is built on it.** If settler mortality cannot be distinguished from zero as a predictor of institutions once smooth geography is taken out, the instrument is **weak**.
- A weak instrument gives you an estimate whose bias runs back toward OLS and whose standard error cannot be trusted. The 2SLS number stops being informative about α **whatever the second stage looks like**.

Losing the second stage would say the effect is imprecise. Losing the first stage says the design is not identifying the thing it claimed to identify.

What this does and does not mean

“The fact that a treatment is insignificant at conventional levels does not necessarily mean that an effect is absent. It may simply not be possible to tie its magnitude down with any precision given the heavily trending and autocorrelated observations that are available. The legacies of history may be as overwhelming as their proponents insist, or as trivial as most historians suspect — we just cannot tell with the data at hand.”

- This is a claim about **what these data can establish**. Colonial history may have had enormous lasting effects, and these designs would still not show them.
- They re-run **one** regression per paper and insist they are not validating or disproving anybody.
- There is a second finding worth as much as the first. In two thirds of the studies at least one region's estimate has the **opposite sign** from the full-sample estimate. In one, every region does.

Where this leaves us

How to read a persistence paper

Six questions. These are what I will ask you in the oral exams.

1. **Assignment.** Who decided which places got treated, and why?
2. **Exclusion.** State in one sentence what must be true for the design to work. If you cannot, you have not understood the paper.
3. **Comparison.** Who are the treated compared to, and how far away are they on the map?
4. **Inference.** Do the standard errors know the data are spatial? How many genuinely independent observations are there?
5. **Mechanism.** Was the channel of persistence measured, or asserted?
6. **Magnitude.** How much of the thing we are trying to explain does this explain?

Four and five are the ones this literature keeps failing. Photograph this.

Where that leaves the week

- Colonial history handed the institutions literature its best natural experiments, and those two papers are as good as the genre gets. The designs are ingenious and the shoe leather is real.
- Conley and Kelly deliver that literature's Freedman moment. Their claim is that the confidence was manufactured by an inference method that did not fit the data.
- Both belong to the same ethic. **The credibility of a finding lives in the design and in the honesty of the standard error.**

The interesting question now is where institutions come from — which is next week.

Next week

- **Week 4 (21 September): Institutions, Culture, and Antecedents.**
- Federalist No. 10 — Madison designing institutions on purpose, knowing exactly what he is trying to prevent.
- Johnson and Koyama on state capacity and constraints. Lamoreaux on the mystery of property rights. Bazzi, Fiszbein and Gebresilasse on frontier culture.

Today asked whether institutions matter and how badly we can measure it. Next week asks where they come from in the first place — and whether culture and institutions are even separable.